

SOLAR COLD STORAGE: Green Energy Can Reduce Crop Wastage

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To overcome the challenges of crop spoilage and high operating costs, solar cold storages are the most effective solution today

Crop spoilage in farms is a major obstacle that India faces. Erratic supply of electricity causes electricity-powered cold storages to remain inactive and storage conditions inside to not remain in the optimal range. On the other hand, diesel generator-driven storage units come with challenges of their own. To overcome these odds and prevent losses, solar-powered cold storages are the most effective solution today.

Challenges that solar storage addresses

Annually, 30 per cent crops, on an average, perish in India due to lack of proper cold storage facilities. Ram Bheke, business development manager, Promethean Power Systems, an India-based cold storage solution provider, says, "Traditional cold storages are associated with high operating costs. For instance, running a complete storage system on diesel has huge expenses associated with it, in the form of price of fuel, maintenance and other associated costs. Moreover, such systems are harmful for the environment."

Reports suggest that these traditional setups increase the total running expenses by 28 per cent in addition to high instances of crop spoilage. Altogether, the producers face major financial losses.

A solar-powered cold storage by ColdHubs (Credit: solutions.sustainia.me)

Solar power comes to the aid here. Emeka Ekeocha, officer at Nigeria-based ColdHubs Ltd, says, "Solar-driven cold rooms in farm clusters and markets help store the crops for much longer. These do not rely on the availability of electricity or exhaust diesel to run. This reduces the running costs of the systems and gives farmers a substantial increase in income."

Benefits of solar storages

The biggest benefit of a solar cold storage unit is reduced dependency on grid supply or diesel. India's abundant solar energy ensures that sufficient power can be saved in the battery or storage units associated with the setup, which can be used in low-light days without hiccups (like monsoons). Alternatively, it cuts down on the various costs associated with diesel generators. As a result, producers using traditional systems can reduce their operational costs by at least 20 per cent.

Post-harvest food spoilage rate is reduced by 80 per cent. Bheke says, "Among the different varieties of produce, leafy vegetables and fruits must be stored carefully. Such items with high water content can get easily spoiled otherwise. With consistent power, solar cold storages can increase the crop shelf-life by up to 10 days. Farmers can store their vegetables for longer until they get the right price." This gives crop producers an increased income of about 25 per cent.

Alongside these, major collateral benefits for the environment—like massive reduction of carbon footprint and reduction of non-renewable resource depletion—are also achieved.

Bheke adds, "Farm owners can find alternate source of incomes as well. We know quite a few who have invested in large-capacity cold storages and rented out portions to other farmers from the area. This acts as an additional income while farm owners can get a payback on their investment quicker. Income through net metering is another option."

The setup

The setup starts with a walk-in storage room, sized according to requirement, with various

